

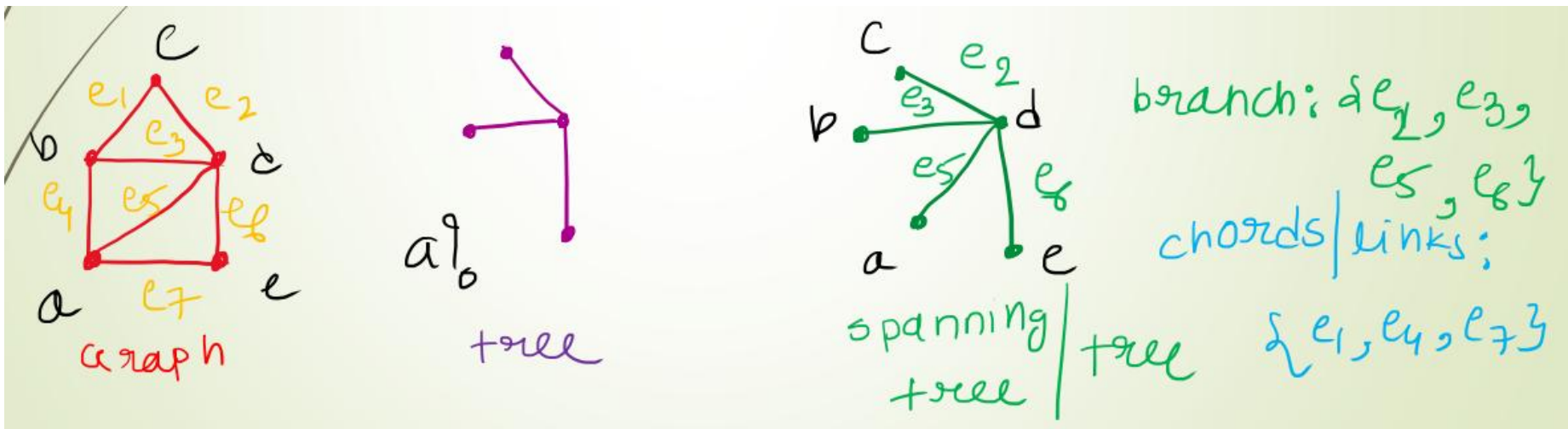
Trees and Cut Set

Outline

- Introduction
- Properties of a tree
- Characteristics of a tree
- Rooted trees
- Path lengths in rooted trees
- Prefix codes
- Spanning trees and Cut-sets
- Transport networks

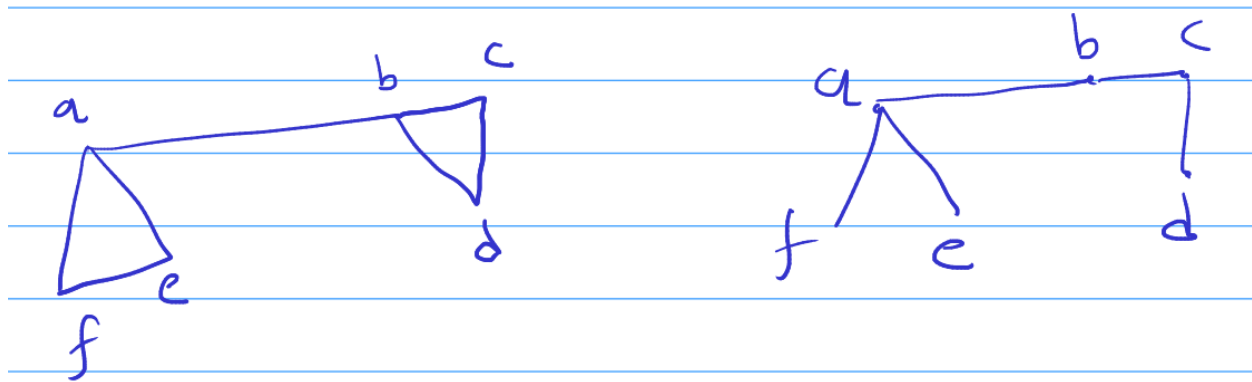
Spanning Trees

- A tree of a graph is a subgraph of the graph which is a tree.
- A spanning tree of a connected graph is a spanning subgraph of the graph which is a tree.
- A branch of a tree is an edge of the graph that is in the tree.
- A chord, or a link of a tree is an edge that is not in the tree.
- The set of chords of a tree is referred to as the complement of the tree.



Spanning Trees

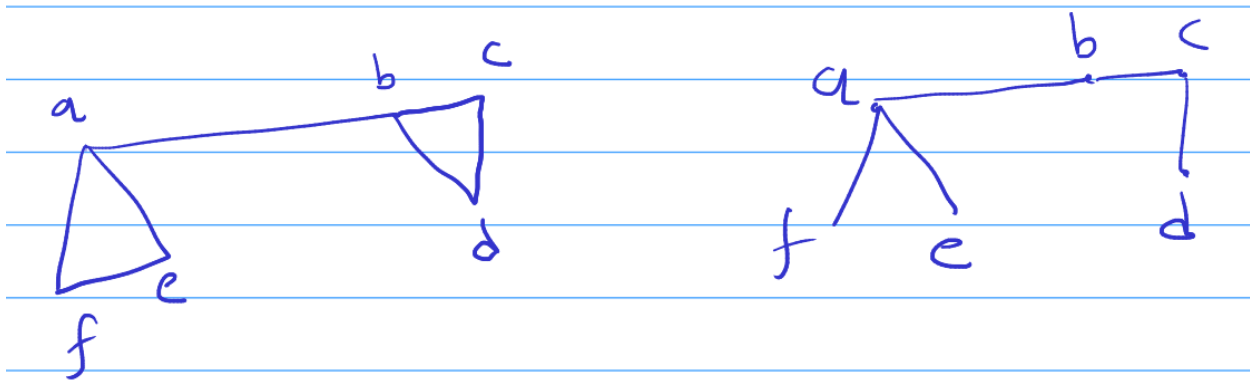
- A connected graph always contains a spanning tree.



- In a connected graph with e edges and v vertices, there are _____ branches in any spanning tree. It follows that relative to any spanning tree, there are _____ chords.

Spanning Trees

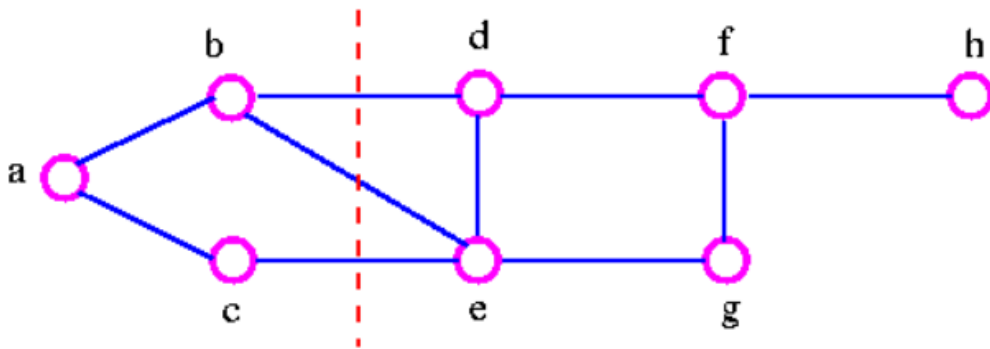
- A connected graph always contains a spanning tree.



- In a connected graph with e edges and v vertices, there are $v-1$ branches in any spanning tree. It follows that relative to any spanning tree, there are $e-v+1$ chords.

Cut-Sets

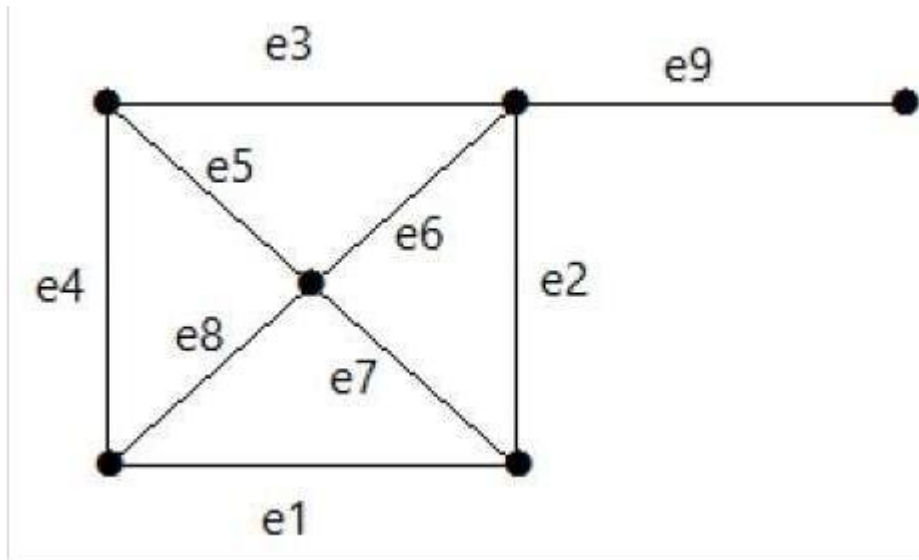
- A cut set of a connected graph G is a set S of edges with the following properties
 - The removal of all edges in S disconnects G .
 - The removal of some (but not all) of edges in S does not disconnect G .



Cut set :

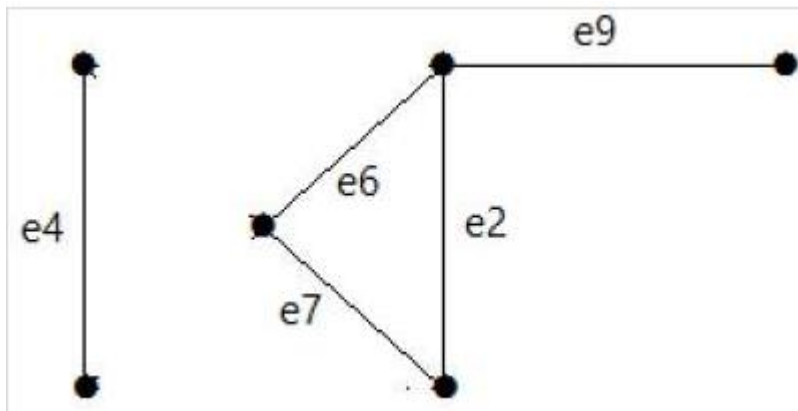
{bd, be, ce }

Cut-Sets



Cut set

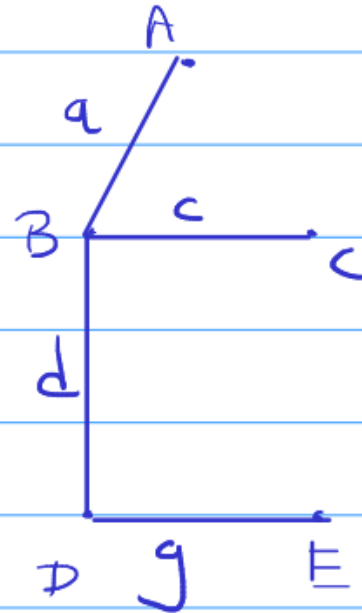
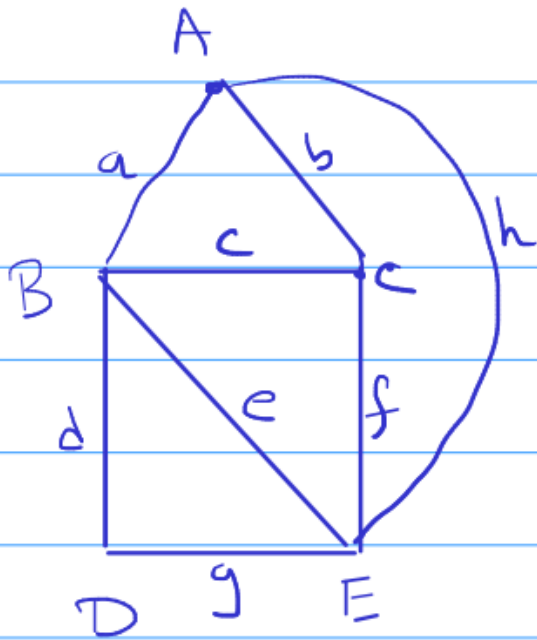
{ e1 , e3, e5, e8 }



Fundamental Circuit

- Let T be a spanning tree in a connected graph G . When a chord is added to a spanning tree T then it forms exactly one circuit. Such a circuit is called a fundamental circuit.

Fundamental Circuit

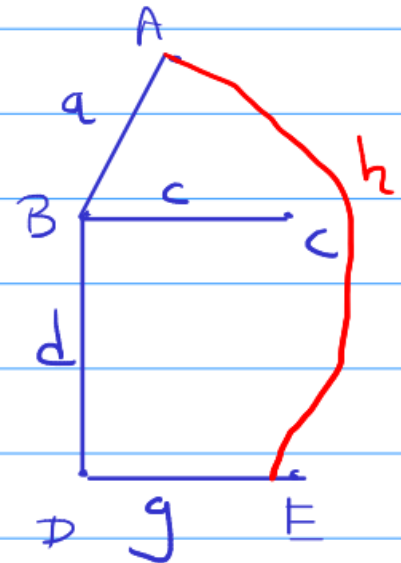
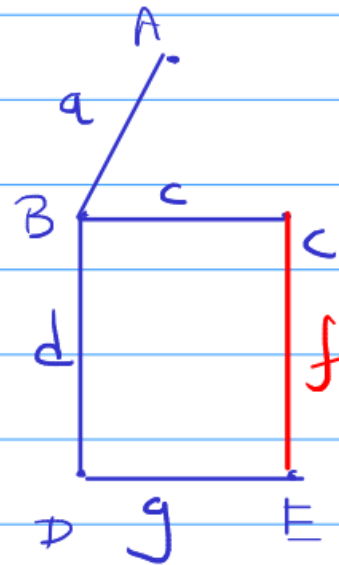
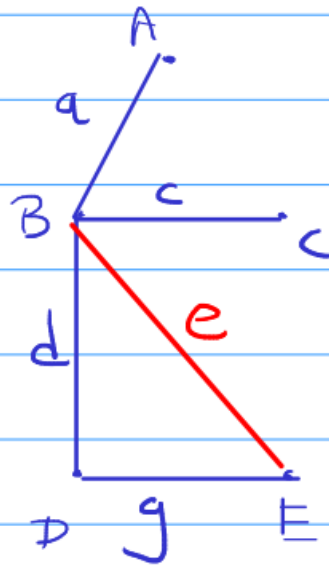
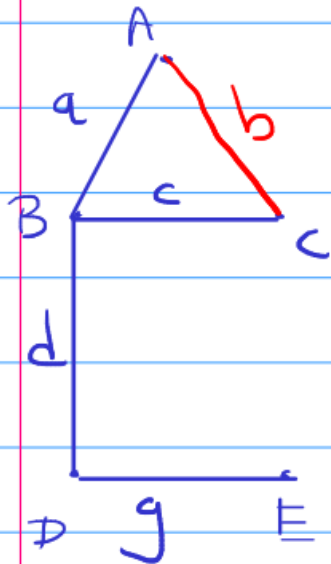


$$\text{Branch} = \{a, c, d, g\}$$

$$\text{Chord} = \{b, e, f, h\}$$

Fundamental Circuits

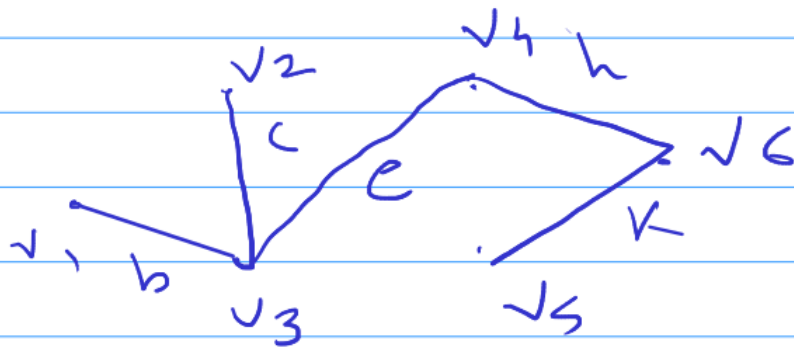
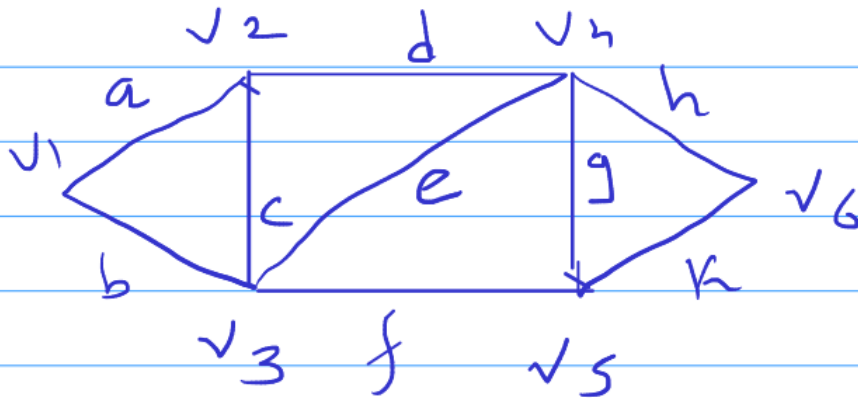
Chords
= {b, e, f, h}



Fundamental cut set

- Let T be a spanning tree of a connected graph G . Then a cutset formed by exactly one branch, say b , of T and possibly some more chords of T is called a Fundamental cut-set of G relative to the spanning tree T .

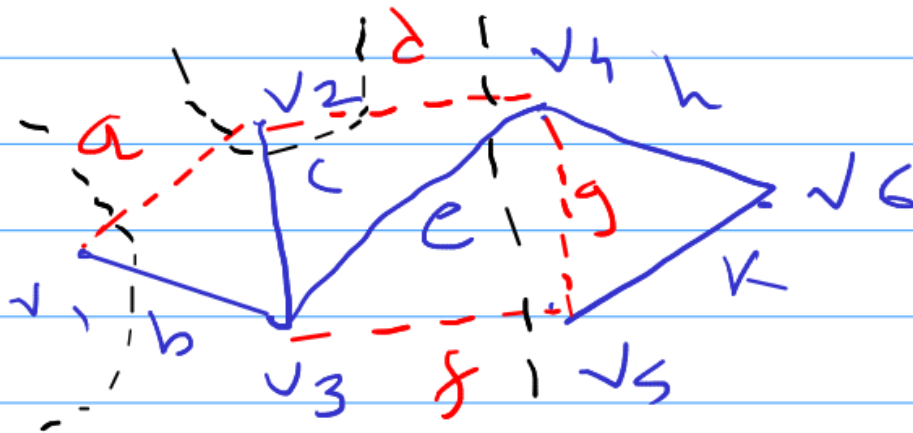
Graph and Corresponding Spanning Tree



$$\text{branch} = \{b, c, e, h, i\}$$

$$\text{chords} = \{a, d, f, g\}$$

Fundamental Cut-Sets



$$C_1 = \{a, b\}$$

$$C_2 = \{a, c, d\}$$

$$C_3 = \{d, e, f\}$$

$$C_4 = \{h, g, f\}$$

$$C_5 = \{h, g, k\}$$

Theorem 5.1

- A circuit and the complement of any spanning tree must have at least one edge in common.

PROOF If there is a circuit that has no common edge with the complement of a spanning tree, the circuit is contained in the spanning tree. However, this is impossible as a tree cannot contain a circuit. □

Theorem 5.2

- A cut set and any spanning tree must have at least one edge in common.

PROOF If there is a cut-set that has no common edge with a spanning tree, the removal of the cut-set will leave the spanning tree intact. However, this means that the removal of the cut-set will not separate the graph into two components, which is in contradiction to the definition of a cut-set. $\square$

Theorem 5.3

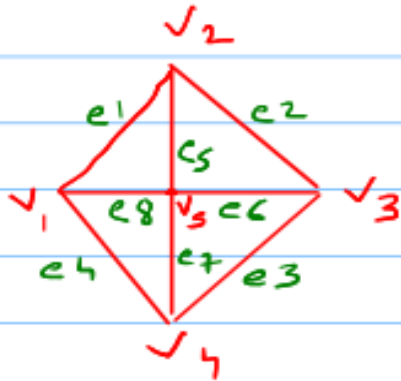
- Every circuit has an even number of edges in common with every cut-set.

PROOF Corresponding to a cut-set, there is a division of the vertices of the graph into two subsets, which are the two sets of vertices in the two components of the graph when the edges in the cut-set are removed. Therefore, a path connecting two vertices in one subset must traverse the edges in the cut-set an even number of times, as illustrated in Fig. 6.18. (The edges in the circuit are drawn in heavy lines.) Since a circuit is a path from some vertex to itself, the theorem follows. $\square$

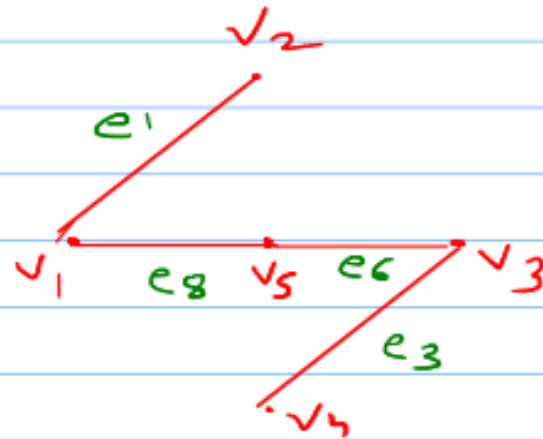
Theorem 5.4

For a given spanning tree, let $D = \{e_1, e_2, e_3, \dots, e_k\}$ be a fundamental cut-set in which e_1 is a branch and $e_2, e_3, \dots, e_k$ are chords of the spanning tree. Then, e_1 is contained in the fundamental circuits corresponding to e_i for $i = 2, 3, \dots, k$. Moreover, e_1 is not contained in any other fundamental circuits.

PROOF Let C be the fundamental circuit corresponding to the chord e_2 . Note that e_2 is in both C and D . Since C and D have an even number of edges in common and e_1 is the only other edge that can possibly be in both C and D ,† e_1 must be contained in C . A similar argument can be applied to the fundamental circuits corresponding to the chords $e_3, e_4, \dots, e_k$. On the other hand, let C' be the fundamental circuit corresponding to any chord not in D . C' cannot contain e_1 , because otherwise, C' and D will have e_1 as the only common edge. □



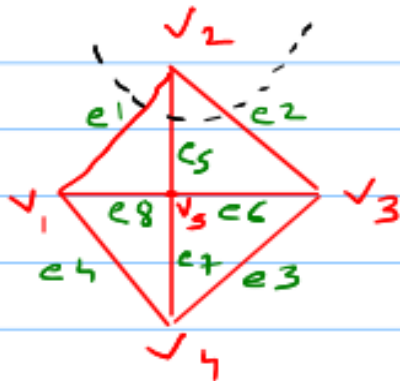
Original Graph



Spanning Tree

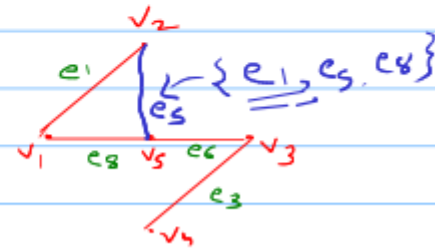
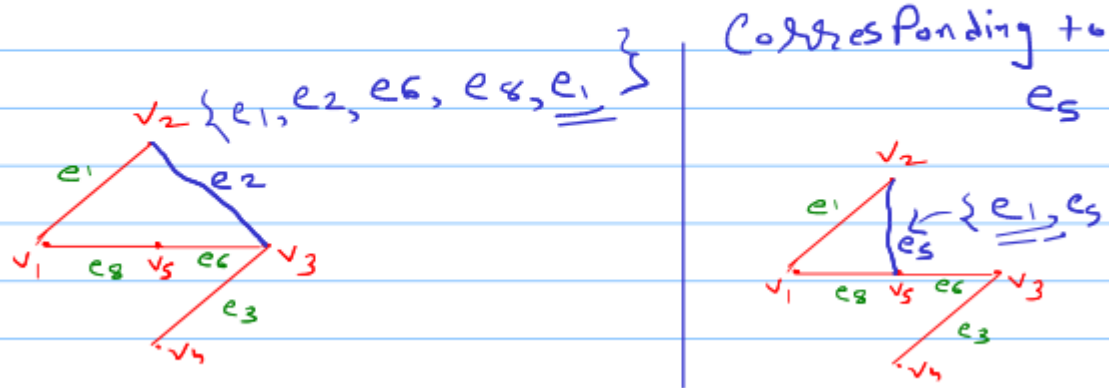
Chords = $\{e_2, e_4, e_5, e_7\}$

branches = $\{e_1, e_8, e_6, e_3\}$

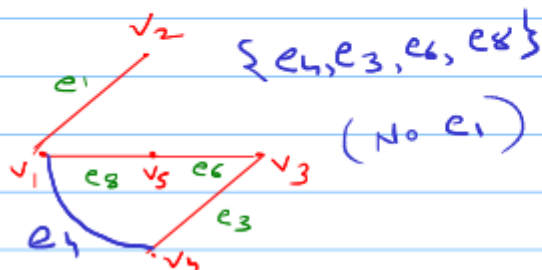


Let $D = \{e_1, e_5, e_2\}$
 (fundamental cut set) Branch Chords

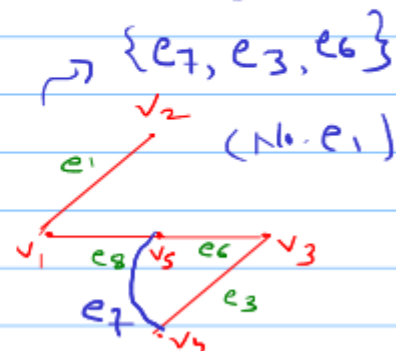
~ fundamental Circuit Corresponding to e_2



Corresponding to e_4



Corresponding to e_7



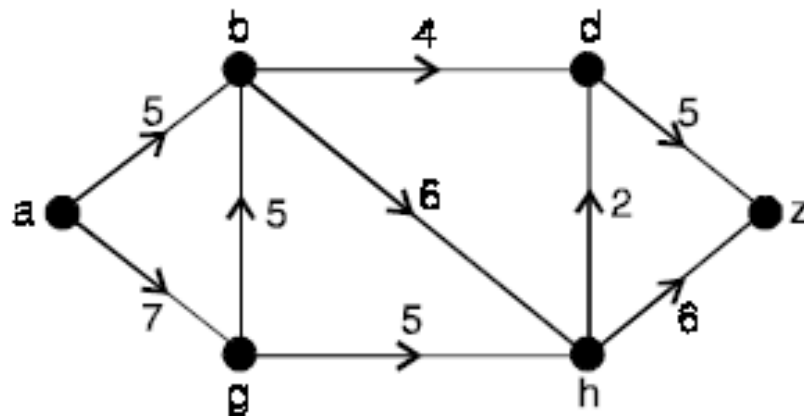
Theorem 5.5

For a given spanning tree, let $C = \{e_1, e_2, e_3, \dots, e_k\}$ be a fundamental circuit in which e_1 is a chord and $e_2, e_3, \dots, e_k$ are branches of the spanning tree. Then, e_1 is contained in the fundamental cut-sets corresponding to e_i for $i = 2, 3, \dots, k$. Moreover, e_1 is not contained in any other fundamental cut-set.

Transport Networks

A weighted directed graph is said to be a *transport network* if the following conditions are satisfied:

1. It is connected and contains no loops.
2. There is one and only one vertex in the graph that has no incoming edge.
3. There is one and only one vertex in the graph that has no outgoing edge.
4. The weight of each edge is a nonnegative real number.



Transport Networks

- In a transport network, the vertex that has no incoming edge is called the source and is denoted by a , the vertex that has no outgoing edge is called the sink and is denoted by z .
- The weight of an edge is called the capacity of the edge. The capacity of the edge (i, j) is denoted by $w(i, j)$.